

Agile maturity model: analysing agile maturity characteristics from the SPICE perspective

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ABSTRACT

The paper discusses structure, quality and content of the currently available agile maturity models. It presents two approaches on how to deal with such models. As a first step of the analysis, the paper contains a compilation of maturity level naming used by these agile maturity models because the variety of level naming is a sign of the variety of understanding and of definition of agile maturity.

While many papers deal with agile from an inside perspective, this paper is written from the perspective of Software Process Improvement (SPI) and Capability Determination, the European Certification and Qualification Association PI Manager Certification Scheme and also the SPI Manifesto.

The paper does not explicitly present its own agile maturity model.

The analysis approaches presented in the paper show that the currently available agile maturity models are not fit for industrial use. The synthesis of an acceptable model seems to be feasible as the analysed models address typical organisational processes including life cycle management. Copyright © 2013 John Wiley & Sons, Ltd.

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1. APPROACH AND METHOD

The following key questions were the starting points for analysing the agile maturity issue:

1. Is an agile maturity model something of relevance?
2. Does a common accepted agile maturity model exist?
3. How would such a model map to Capability Maturity Model Integration (CMMI)?
4. Does this model fit to common accepted standards as stated in ISO/IEC 15504 Part 2?
5. How would such a model map to ISO/IEC 15504 Part 5:2012?
6. If such a model could be found, would it more be like a stairway or more like a spider web?

For the first question, a survey was performed. Considering that agile maturity is of high interest, the results were published in 2012 [1].

To answer the second and third questions, an internet search was undertaken, and its results were sampled including 40 sources dealing with agile maturity. As completeness criterion, somewhat as a jackknife algorithm was used, assuming that starting with a Google search and following the links in the found documents as well as checking given references, no more sources will be found when a

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source is found the third time following a chain of links. The result contains one cluster of models that somewhat refer to CMMI and other models that follow their own maturity approaches. The detailed preliminary analysis is documented in section 3.

As there was no common accepted agile maturity model found, questions 4 and 5 became more interesting. The approach and its result are described in section 4.

The first answer of question 6 is mentioned in section 5.

As it was clear that this paper did not have the intention to deliver a complete detailed analysis or a complete agile maturity model, section 6 deals with some useful follow up actions.

2. THE CURRENT DISCUSSION ON AGILE MATURITY MODELS

While it is common opinion that it is possible for an agile organisation to reach high CMMI® maturity levels or high Software Process Improvement and Capability Determination (SPICE) Capability levels [2], it seems that some agile gurus are not happy with the support offered by CMMI or SPICE. As one of the consequences, we have about 40 different models that call themselves agile maturity models.

There are several types of agile maturity models published, mainly in the Internet. There are also some principal thoughts published about agile maturity. The discussion about agile maturity is influenced by ideas of the CMMI® model. See also Schweigert *et al.* [3].

So it seems to be adequate to group the published agile maturity models into those which are close to the level structure of CMMI® [4–11], those which have a level structure at all [5, 9, 12–28] and those which do not use explicit levels [6, 14, 16, 21, 29–41, 61]. A detailed synopsis of level naming can be found in Schweigert *et al.* [42].

All of these models have in mind the approach, which was one of the goals of the BOOTSTRAP methodology about 20 years ago: ‘Change and reorganize the software development and maintenance activities so that the software production as a whole better complies with business needs’ [43]. The methods changed, but the goals remain.

This result creates however a heavy challenge for the SPI community. Like at the early times of SPICE, many models (e.g. ISO 9000, Trillium, TickIT, CMM, BOOTSTRAP, ISO 12207 and ISO 15288) were on the market, and there was no method to make their results comparable.

The need for structured modelling is agreed in the whole IT business (see, e.g. TestSPICE® for the software testing business [44, 45]).

The challenge for SPICE assessors in an agile environment is the heterogeneity of agile implementations. So there is a need to distinguish between malpractice and next practice [46]. To do so, it might be helpful to identify anti-patterns to mark malpractices [47] and also implementation patterns [48]. Looking deeper into the agile maturity issue, it becomes clear that improving agile development and management is not only an issue of formal capability or maturity levels but also an issue of values, emotions and culture [49, 50, 64].

An analysis of the quality of a subset of these agile maturity models was performed by Özcan-Top [51]. Instead of asking about the naming and content, this analysis took a set of model quality criteria (fitness for purpose, completeness, definition of agile levels, objectivity, correctness and consistency) checking if these quality criteria were fully, largely, partially or not fulfilled by the models of Patel [8], Yin [11], Sidky [27], Benefield [18] and Ambler. As an overall result, it could be said that the rating of the analysed subset was more or less poor [51].

3. DO AGILE MATURITY MODELS REALLY MEASURE AGILITY?

The first idea, when analysing agile maturity models in detail, was to check which capability or maturity levels they are supporting. Trying to map the available agile maturity models to capability indicators of SPICE turned out to be very difficult as many authors of agile maturity models have an undisciplined wording. So there is no clear connotation if a model describing a level, a process attribute, a process, an outcome or an indicator is like a base practice. Even a rough analysis shows that none of the analysed agile maturity models fulfil the requirements of ISO/IEC 15504 Part 2 [60].

So, typically, it is very hard (sometimes also impossible) to develop a direct mapping between agile maturity models and SPICE capability levels.

During the 2012 EuroSPI in Vienna a team of experts tried to develop a first analysis of the content of the above mentioned agile maturity models. The result is aggregated in Figure 1.

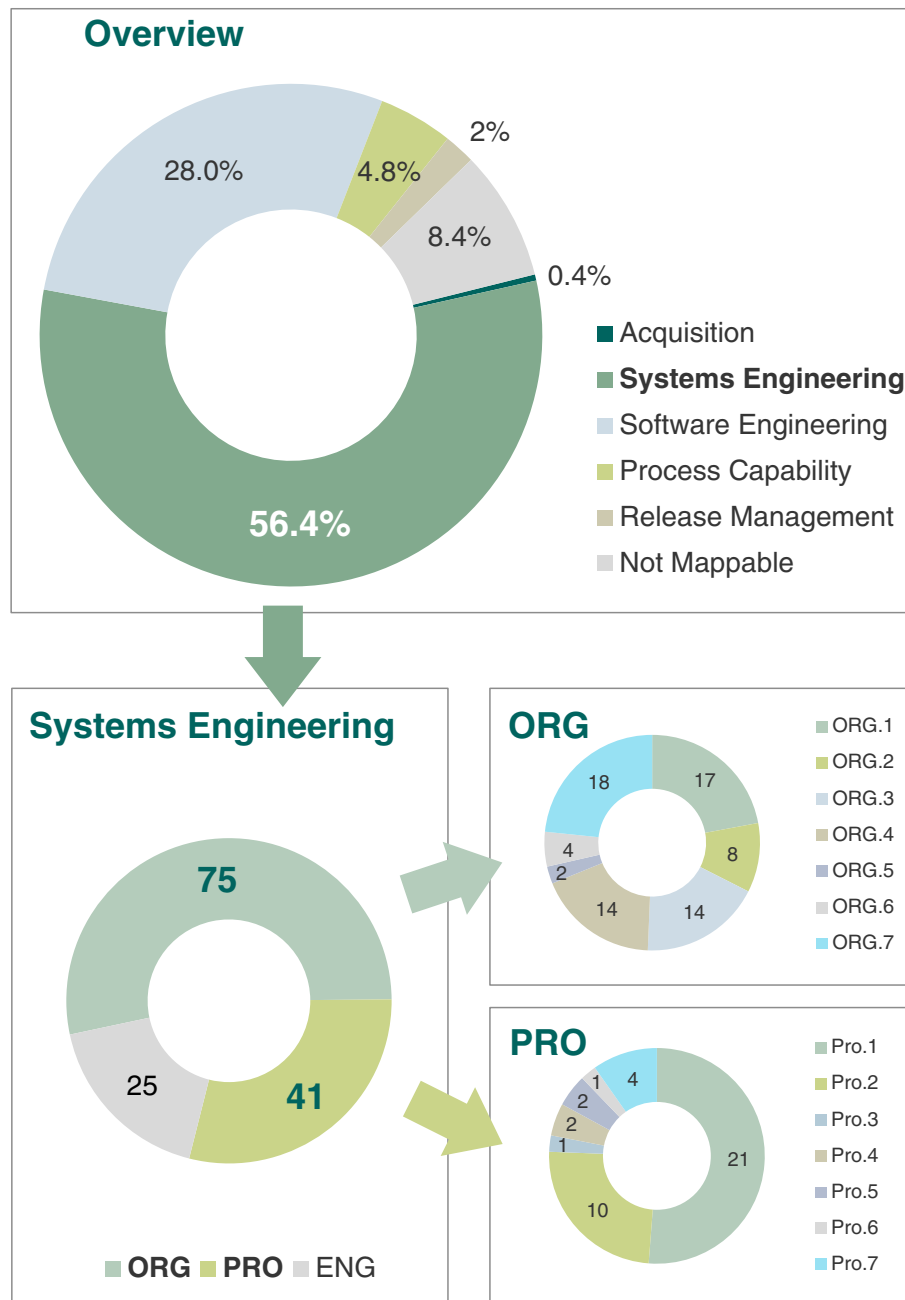


Figure 1. Results of the 2012 EuroSPI Assessor Workshop in Vienna.

How was this result developed?

The workshop was prepared by atomizing a sample of agile maturity models in order to check what they really contain. Using a sample of 12 of the 40 agile maturity models and extracting atomized characteristics out of these models, we synthesized a set of 600 atomized characteristics.

For each characteristic, the following attributes are applied:

- ID
- Author
- Type: {value, principle, process, purpose, outcome, practice, work product, process attribute and level}
- Article
- Content (the extracted statement)

These attributes were placed in an analysis card (A5). About 600 of these cards were prepared for the agile maturity workshop at the 2012 EuroSPI Conference.

Sometimes, it was not possible to define the exact type of the statement. In this case, more than one type was chosen.

This approach was pragmatic. There exist more sophisticated approaches [52], and the above one proved, however, to be effective for the development of a first sample.

During a workshop that took place at the EuroSPI 2012 in Vienna, a subset of 250 characteristics of the sample was mapped by the participants to ISO/IEC 15504 Part 5 2012 [53]. The detailed result is as follows:

Most important processes:

ORG.1 Life cycle model management

ORG.3 Project portfolio management

ORG.4 Human resource management

ORG.7 Organization management

PRO.1 Project planning

PRO.2 Project assessment and control

It was also found that core software engineering practices were highly relevant.

Even if it was not possible to map the agile maturity characteristics directly to capability levels, it should be taken into account that the processes mentioned earlier are supporting processes for levels 2 and 3.

If maturity is an organizational topic, which was especially highlighted by Hammer [54], it is reasonable that some agile maturity models deal either with maturity characteristics or with organizational processes.

But also often agile maturity models deal with simple software development practices, as shown in the figure above.

These results lead to the conclusion that agile maturity has a view different from the classic one regarding capability; however, its approach to process/practice implementation of organizational or project management processes follows the agile principles.

More research is also important regarding that the definition of agile maturity depends on the definition of agile [59] itself. It was reported that this definition is also evolving [55]. At the very end, it must also be taken into consideration that agility is definitely not the ultimate goal but one important aspect of high performance teams [56, 65].

4. GENERAL INTERPRETATION

Considering the aforementioned results, it is certainly not surprising that several related initiatives were recently born.

One of these is of [57], consolidating the values and principles that a group of experts within the EuroSPI (<http://www.eurospi.net/>) community proposed to be of key significance. The three key values of the SPI Manifesto are that SPI

- ‘must involve people actively and affect their daily activities’,
- ‘is what you do to make business successful’ and
- ‘is inherently linked with change’.

The European Certification and Qualification Association Software Systems SPI Manager Certification Scheme developed by the authors of this paper, among others, carefully covers these issues [62, 63].

Software Engineering Method and Theory (SEMAT) [58] is an initiative with a high number of supporters who agree that

Software engineering is gravely hampered today by immature practices. Specific problems include:

- The prevalence of fads more typical of fashion industry than of an engineering discipline.
- The lack of a sound, widely accepted theoretical basis.
- The huge number of methods and method variants, with differences little understood and artificially magnified.
- The lack of credible experimental evaluation and validation.
- The split between industry practice and academic research.

The survey results reported in this paper are intimately related to the aforementioned initiatives and can be considered to validate the general interpreting ideas advanced in the following paragraphs.

It is a general fact that a sound theoretical basis is usually built on mathematical approaches, which are clearly used in all widely accepted sciences. Let us use analogies from mathematics to characterize the state-of-the art in process improvement methods.

In mathematics, there can be many interesting statements (usually called theorems, lemmas, etc.), which are deductively proven to be true (valid) starting from axioms whose truth is taken for granted within the particular domain of analysis. Still, there are mostly many ways to get from the axioms to the same theorem. These ways may be very different as far as their elegance (aesthetics), effectiveness and usability are concerned, which have a definite impact on their capability to extend and generalize the results.

Abstraction, being essential for model building, is a key process in mathematics just as in software engineering or software process modelling. From the most commonly used mathematical abstraction called 'natural number', to the highly different abstractions used in classical analysis and topology, all of mathematics shows how the different models applied in given theories can lead to a more or less elegant and extensible approach to the same or even revolutionarily new concepts.

It is a similar situation we are facing in SPI. We have many different approaches and statements that are claimed to be effective in leading to the achievement of the same business goals. Nevertheless, SPI is not rocket science, so the deductive proof of these claims is mostly impossible; by consequence, they must be and are usually proven empirically, which means inductive reasoning. The reference to best practices in either agile or heavyweight approaches claimed to be generally effective is clearly inductive reasoning. This is definitely part of the common theoretical foundation of process improvement whether agile or heavyweight.

The key analogy with mathematics is that there are different ways (CMMI, SPICE, agile maturity models as discussed in the survey, etc.) to get to the conclusion that a method is effective in leading to the achievement of business objectives. These ways are just as different in their elegance (aesthetics) and usability as there are very different approaches in mathematics to the same concepts. It is also true that elegance (aesthetics) depends on subjective taste, while effectiveness and usability may be objectively measurable. By consequence, there will always be differences in the opinions regarding the elegance of approaches while at the same time all parties may measurably support their claims regarding the effectiveness or usability of their approach.

At this point, we have to recur to one of the key principles applied by SEMAT as well: separation of concerns. Subjective concerns such as elegance (aesthetics), for example, should be separated from objectively measurable concerns such as effectiveness and usability, especially if the objective measurements are not differentiating enough. And we are at the heart of the success factors addressed in [58]: better, faster and happier.

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